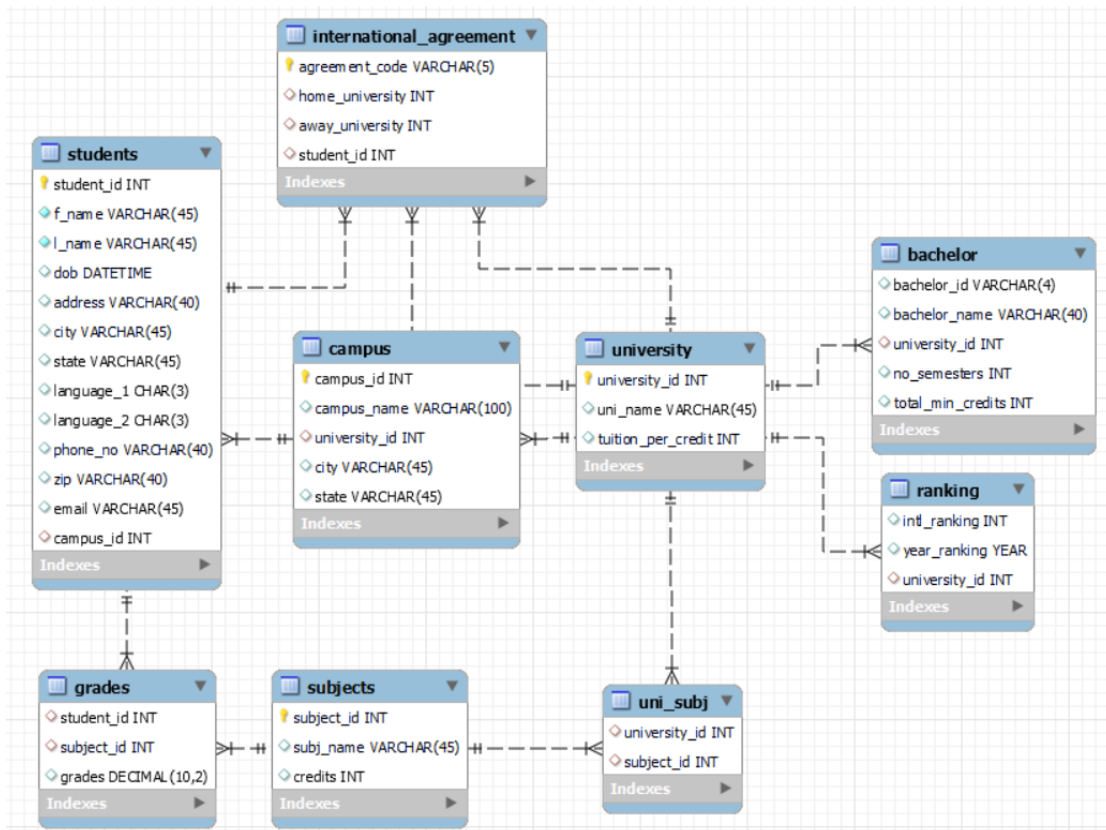


ERASMUS MINI PROJECT

Using the erasmus database, answer the following questions. The database has the following structure:



- What is the average age of students who have outstanding grades? Provide the table with EXCELLENT if they have a 9 or 10, GOOD if they have 7 or 8, PASS if they have a 5 or 6, and FAIL if it is less than 5.

	Grade_Category	Average_Age
▶	FAIL	27
	PASS	27
	EXCELLENT	28
	GOOD	28

- What is the average age of students by university?

	university_name	average_age
	University of Florida	43
	University of Virginia	37
	Massachusetts Institute of Technology	35
	University of North Carolina at Chapel Hill	31
	University of Chicago	30
	University of Illinois Urbana-Champaign	29
	Columbia University	29
	University of Pennsylvania	29
▶	University of Georgia	28
	University of Colorado Boulder	28
	University of Wisconsin-Madison	28
	Harvard University	27
	University of Michigan, Ann Arbor	26
	Indiana University Bloomington	26
	University of California, San Diego	26
	University of California, Los Angeles	25
	Northwestern University	25
	University of Texas at Austin	24
	Vanderbilt University	24
	Johns Hopkins University	23
	Emory University	23
	University of Washington	23
	Rice University	22
	University of Minnesota, Twin Cities	22

- What is the proportion of students who failed each subject? Provide the subject name, number of students who failed, total number of students, and the proportion of students who failed (as a percentage) for each subject. Display the results in descending order of the proportion of students who failed.

	Subject_Name	Number_of_Failures	Number_of_Students	Proportion_of_Failures
▶	Renewable Energy Engineering	7	8	87.50%
	Social Psychology	12	15	80.00%
	Film Production	7	9	77.78%
	English Literature	10	13	76.92%
	Astrobiology	9	12	75.00%
	Sociology	6	8	75.00%
	Neuroscience	11	15	73.33%
	Biochemistry	8	11	72.73%
	Mathematics	7	10	70.00%
	Seismic Interpreter	8	12	66.67%
	Renewable Energy	8	12	66.67%
	Physics	6	9	66.67%
	Comparative Literature	8	12	66.67%
	Fashion Design	11	17	64.71%
	Anthropology	7	11	63.64%
	Economics	7	11	63.64%
	Geology	5	8	62.50%
	Graphic Design	10	16	62.50%
	Nanotechnology	11	18	61.11%
	Computer Science	6	10	60.00%
	Chemistry	6	10	60.00%
	Geographic Information Systems	7	12	58.33%
	Sports Science	11	19	57.89%
	Astronautics	8	14	57.14%
	Biotechnology	8	14	57.14%

- What is the average grade of students who have done an erasmus compared to those who have not?

	Erasmus_Status	AVG_Grade
▶	wo_erasmus	5.036714
	w_erasmus	4.829543

5. For each university, identify the number of bachelor's, master's and PhD degrees awarded. Provide the university ID and name along with the count for each type of degree.

	university_id	uni_name	Bachelor_Count	Master_Count	Phd_Count
▶	1	Harvard University	0	1	0
	2	Stanford University	1	1	1
	3	Massachusetts Institute of Technology	4	1	1
	4	California Institute of Technology	3	0	0
	5	Yale University	0	1	1
	6	Princeton University	3	1	2
	7	Columbia University	1	2	1
	8	University of Chicago	0	2	0
	9	University of Pennsylvania	2	1	1
	10	Johns Hopkins University	1	1	0
	11	Northwestern University	0	2	0
	12	Duke University	2	1	1
	13	Cornell University	2	2	0
	14	Brown University	2	1	1
	15	Rice University	1	1	1
	16	University of Notre Dame	0	2	1
	17	Vanderbilt University	1	0	2

6. Which are the top 5 universities with the highest average ranking over the years? Provide the University ID, University Name, and Average Ranking.

	University_ID	University_Name	Average_Ranking
▶	40	University of California, Davis	70
	33	University of Minnesota, Twin Cities	67
	29	University of Illinois Urbana-Champaign	66
	4	California Institute of Technology	65
	11	Northwestern University	63

7. Provide de id, the name, the last name, the name of the home university and the email of the 10 students that have been on an international agreement more times.

	student_id	f_name	l_name	Home_University	email	Agreement_Count
▶	9	Sage	Wieser	Northwestern University	sage.wieser@example.com	3
	37	Bender	Rodriguez	University of Michigan, Ann Arbor	bender.rodriguez@example.com	3
	41	Harry	Potter	Columbia University	harry.potter@example.com	3
	53	Donkey	Kong	University of Florida	donkey.kong@example.com	3
	45	Bart	Simpson	Columbia University	bartt.simpson@example.com	3
	21	Buddy	Thunderstruck	Georgetown University	buddy.thunder@example.com	2
	56	Hobbit	Frodo	Rice University	hobbit.frodo@example.com	2
	74	Liam	Young	Indiana University Bloomington	liam.young@example.com	2
	62	Michael	Johnson	University of Iowa	michael.johnson@example.com	2
	77	Amelia	Walker	University of California, Los Angeles	amelia.walker@example.com	2

8. Make a query where, by modifying the international agreement number, you can identify the id and name of the student who did the exchange, the name of the home university and the name of the city where the exchange took place.

	agreement_code	student_id	Student_First_Name	Student_Last_Name	Home_University	Away_University
▶	A2B9C	74	Liam	Young	Indiana University Bloomington	University of Arizona
	agreement_code	student_id	Student_First_Name	Student_Last_Name	Home_University	Away_University
▶	90P82	44	Hermione	Granger	University of Virginia	Princeton University

Bonus: Now you can try to use procedures to parameterize the query.

9. Find and display the number of universities offering each subject, along with the average grade for each subject.

	Subject	Average_Grade	Num_Universidades
►	Aerospace Engineering	7	9
	Anthropology	4	8
	Archaeology	5	8
	Artificial Intelligence	6	9
	Astrobiology	4	9
	Astrogeology	4	8
	Astronautics	4	11
	Astronomy	6	15
	Astrophysics	4	9
	Behavioral Economics	6	11
	Biochemistry	4	10
	Biomedical Engineering	6	8
	Biotechnology	5	6
	Business Administration	5	10
	Business Ethics	5	12
	Chemistry	5	7
	Civil Engineering	5	11
	Climate Change Studies	6	8
	Comparative Literature	4	4
	Computer Graphics	5	6
	Computer Science	4	6
	Criminal Justice	5	14
	Criminology	5	10

10. Find the top 5 cities with the highest percentage of students who have outstanding grades (9 or 10). Provide the city, state, and the percentage of outstanding students for each city.

	City	State	Percentage_Outstanding
►	Madison	Wisconsin	30.00
	Evanston	Illinois	20.00
	Athens	Georgia	20.00
	Philadelphia	Pennsylvania	17.50
	Houston	Texas	15.00

11. Compare the universities that send the most students with the universities that receive the most students. Do it in 2 queries.

	uni_name	sent_students		uni_name	received_students
►	Columbia University	8	►	University of California, Davis	5
	University of Michigan, Ann Arbor	6		University of Colorado Boulder	4
	University of California, Davis	6		University of Minnesota, Twin Cities	4
	Georgetown University	5		California Institute of Technology	4
	University of Virginia	5		University of Arizona	4

Bonus: Now you can try to join both queries by using "UNION ALL" operator.

	uni_name	sent_students	received_students
►	Columbia University	8	NULL
	University of Michigan, Ann Arbor	6	NULL
	University of California, Davis	6	NULL
	Georgetown University	5	NULL
	University of Virginia	5	NULL
	University of California, Davis	NULL	5
	University of Colorado Boulder	NULL	4
	University of Minnesota, Twin Cities	NULL	4
	California Institute of Technology	NULL	4
	University of Arizona	NULL	4